Housing Decision Analysis

Option A: Trailer on Parents' Property vs Option B: $98,500 House Purchase

Muscle Shoals, AL | Monte Carlo Simulation (1,000 runs) | Generated April 09, 2026

Simulation Parameters

Parameter	Value	Note
Combined take-home	$5,100/mo	Both jobs, after tax
Starting savings	$22,000	At wedding, Nov 2026
Home price (Option B)	$98,500	
Down payment	$3,448	3.5%
Closing costs	$4,925	5.0%
Total upfront cost	$8,372	
Cash after closing	$13,628	Remaining cushion
Loan amount	$95,052	30-yr, 7.0% fixed
Monthly mortgage	$632.39	Fixed — does not inflate
Option A gas	$600/mo	Long commute, 2 cars
Option B gas	$300/mo	Shorter commute, 2 cars
Rental income (B)	$800–$1000/mo	Starts month 38, 10% vacancy
Home appreciation	3.0%/yr	Conservative
Expense inflation	3.0%/yr	
Invest return on cash	4.5%/yr	HYSA / fund
Simulation length	60 months (5 yrs)	
Monte Carlo runs	1,000	P10/P90 uncertainty bands

Key Results Summary

Metric	Option A (Trailer)	Option B (House)	Winner
Mean net worth — final month (no time value)	$200,557	$195,623	**A**
Mean net worth — final month (with time value)	$162,946	$195,600	**B**
10th percentile net worth (no time value)	$199,822	$193,136	**A**
90th percentile net worth (no time value)	$201,310	$198,177	**A**
Break-even month (median, no time value)	—	Month N/A	**—**
% simulations: B wins within horizon (no TV)	—	0%	**—**
Break-even month (median, with time value)	—	Month 19	**—**
% simulations: B wins within horizon (with TV)	—	100%	**—**
Cumulative gas cost (A vs B)	~$36,000.0	~$18,000.0	**B saves $18,000.0**
Home equity at final month (mean)	—	$24,714	**B**
Mean rental income earned (lifetime)	—	$19,147	**B**

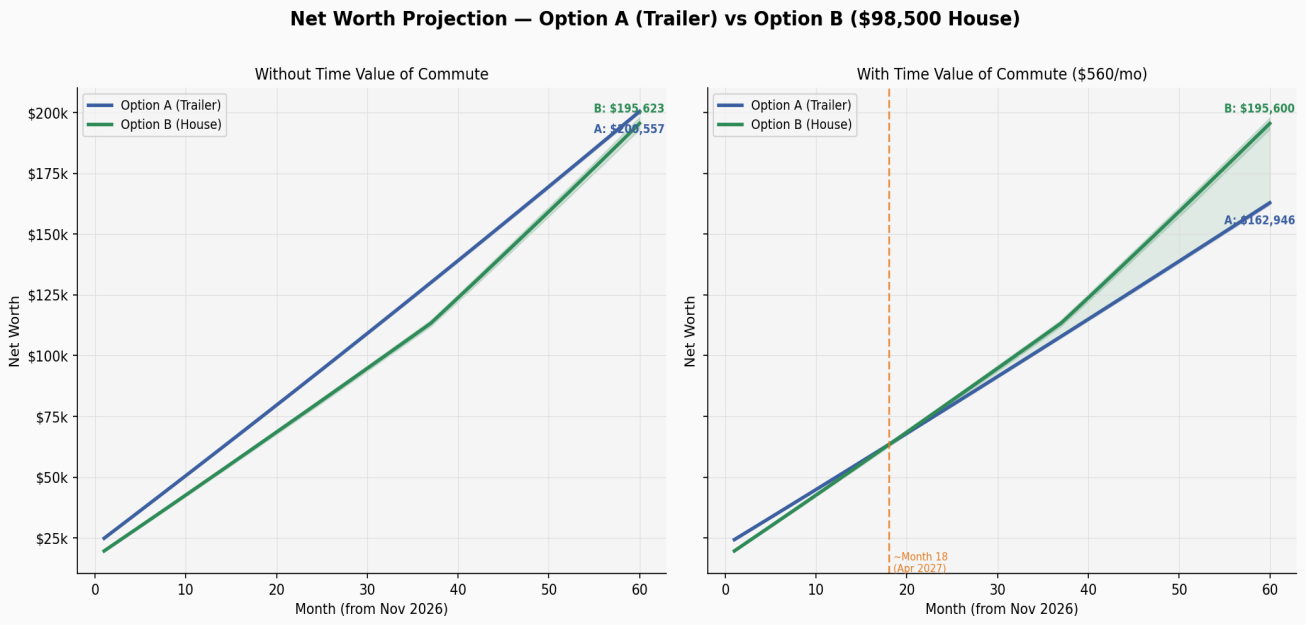

Chart 1: Net Worth Projection — 60-month horizon with Monte Carlo P10–P90 bands. Left: no time value. Right: includes $560/mo commute opportunity cost.

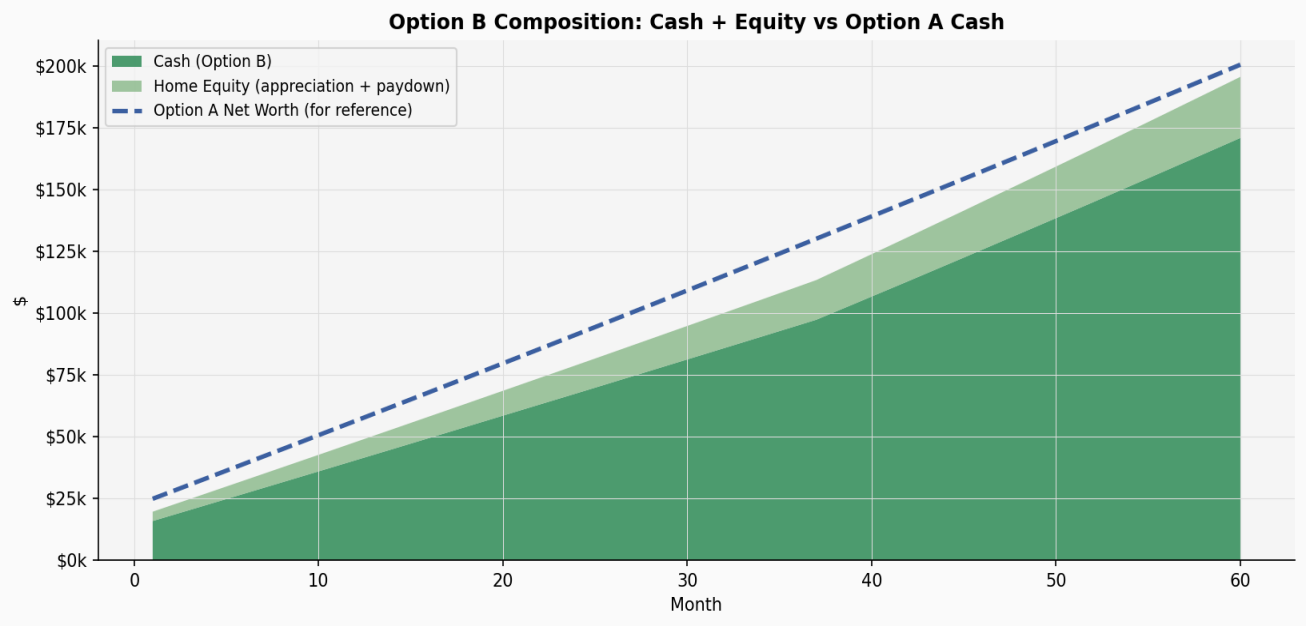

Chart 2: Option B component breakdown — cash balance plus home equity. Option A net worth shown dashed for comparison.

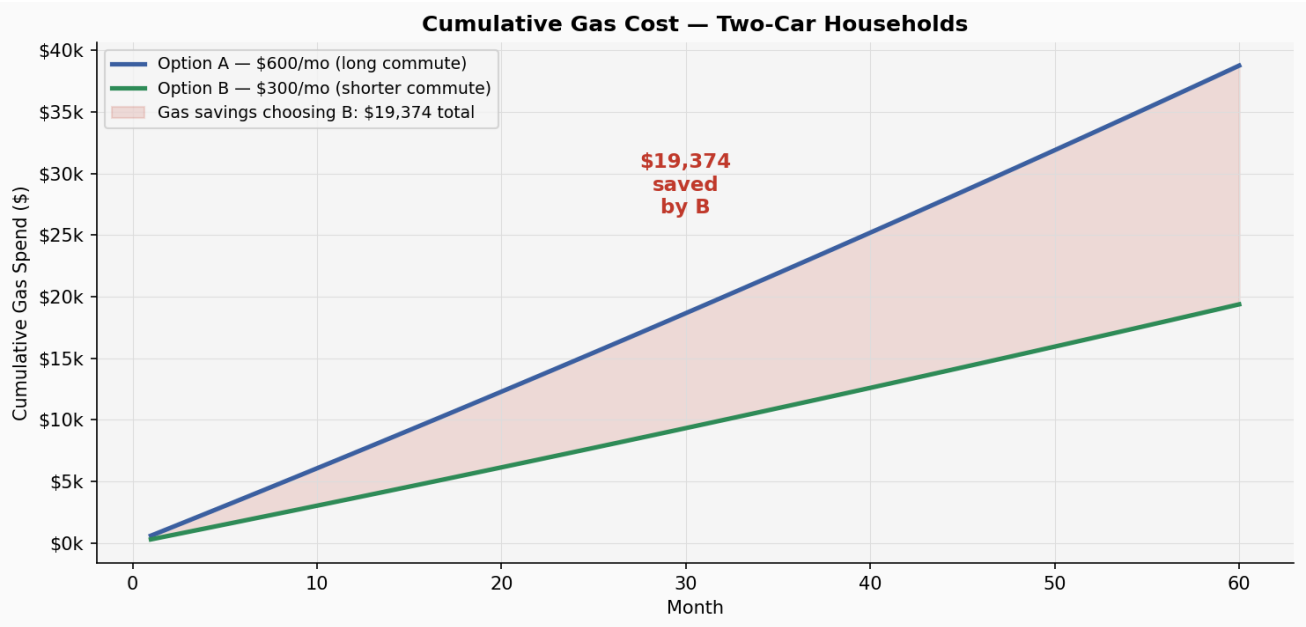

Chart 3: Cumulative gas expenditure. Option A pays $600/mo vs Option B $300/mo — the difference compounds over the full horizon.

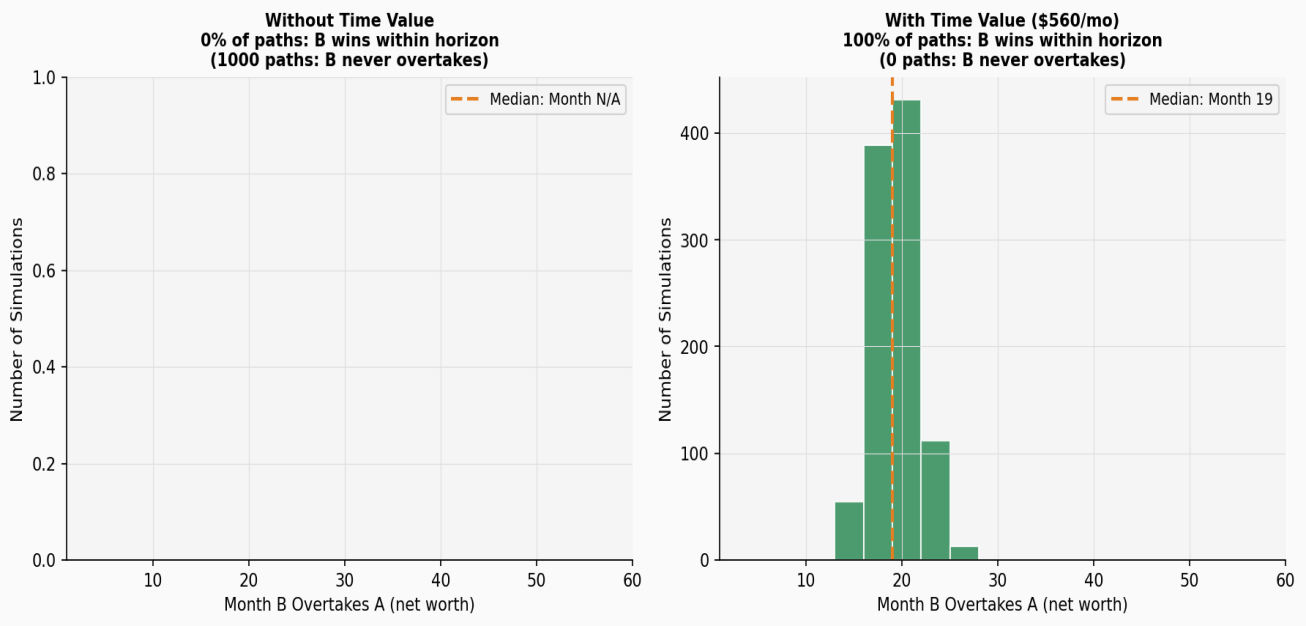

Chart 4: Distribution of break-even month across all simulation paths.

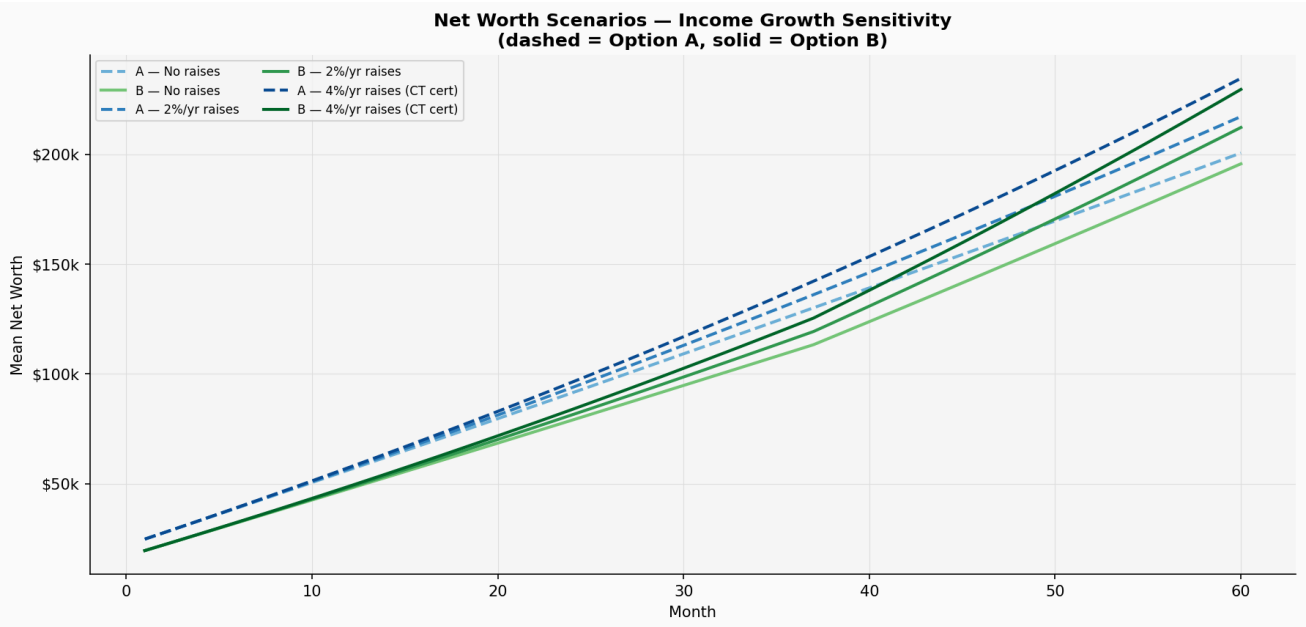

Chart 5: Scenario analysis — net worth trajectories under different income growth assumptions.

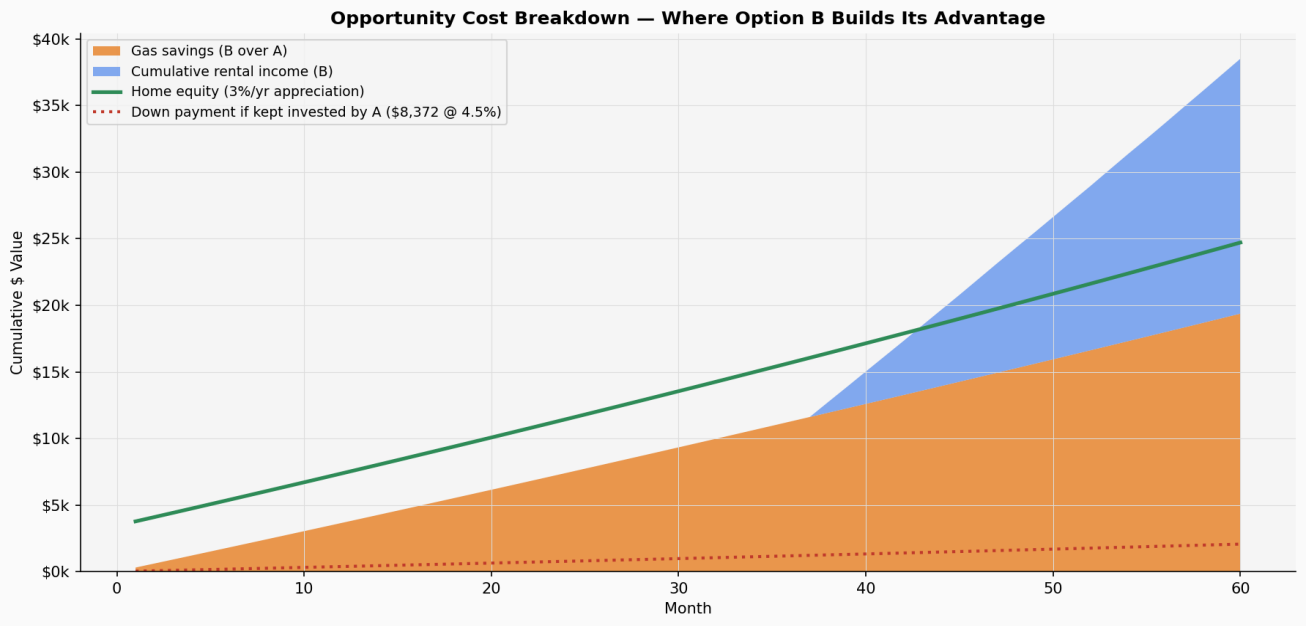

Chart 6: Opportunity cost decomposition — where Option B's long-run advantage originates.

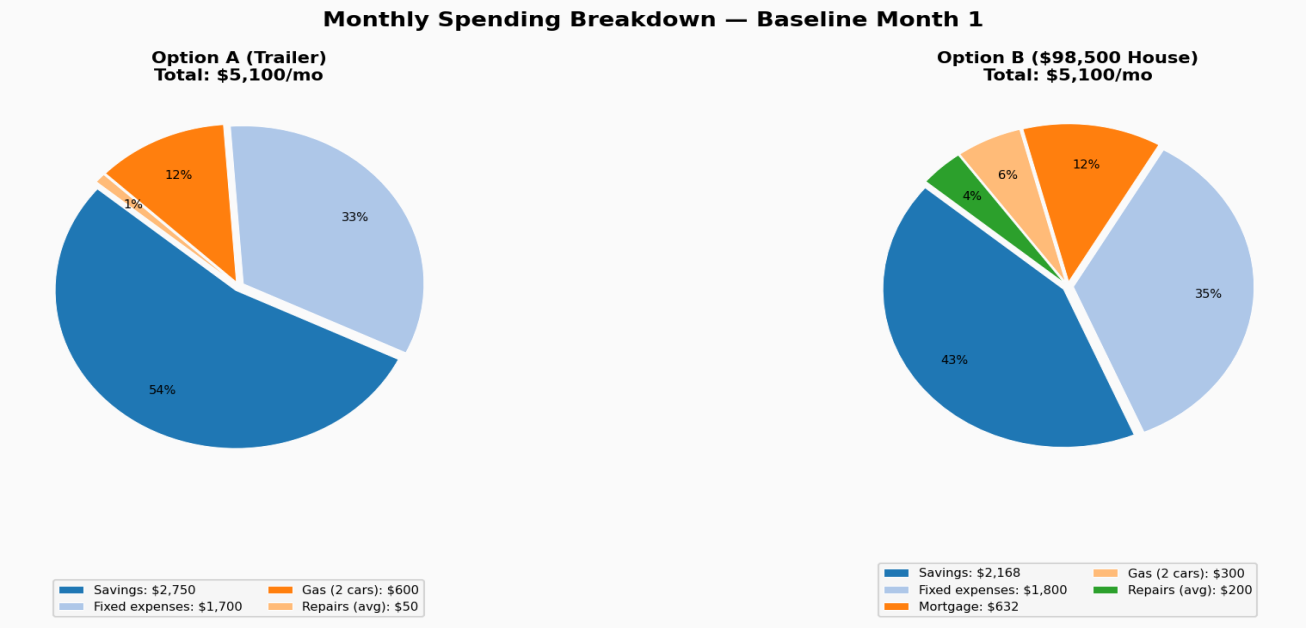

Chart 7: Monthly spending composition using actual config values.

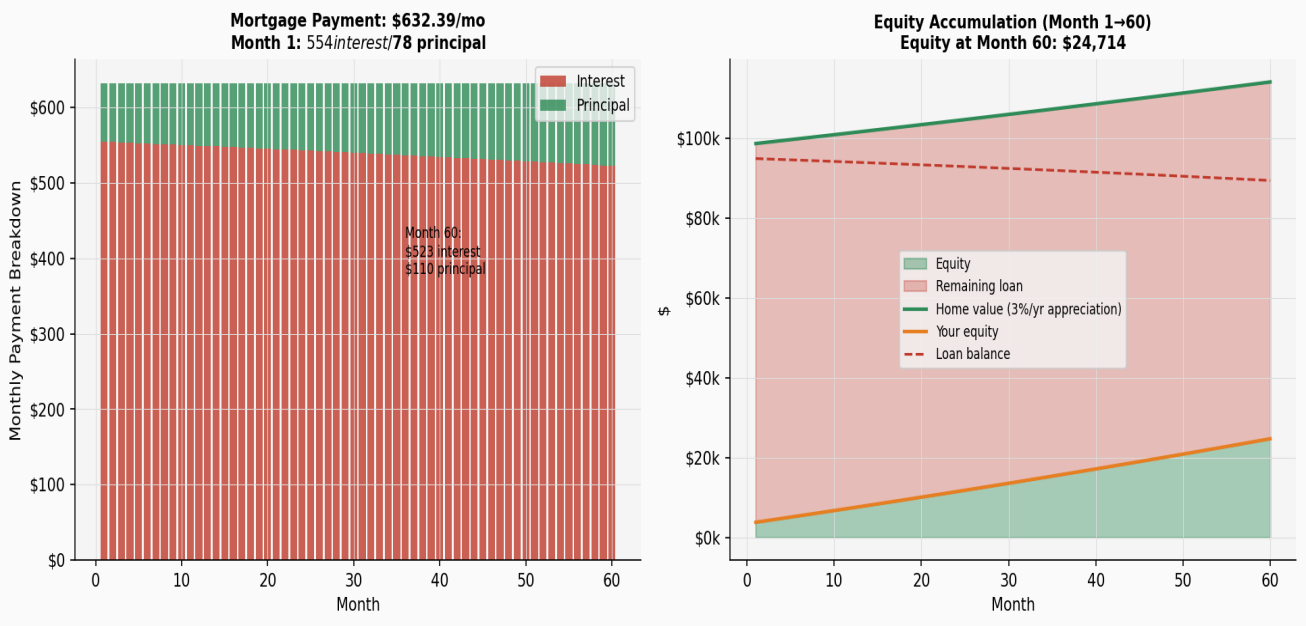

Chart 8: Mortgage amortization detail for $98,500 home at 7.0% over 30 years.

Analyst Verdict

Base scenario (no time value): Option A leads by $4,934 at month 60. 0% of simulation paths show B overtaking A within the 60-month horizon.

With time value of commute ($560/mo): Option B leads by $32,654. 100% of paths show B winning.

Key risk factors to monitor:

- Repair spike in years 1–2 before rental income starts (month 38). Cash cushion after closing: $13,628.
- Vacancy risk on rental (modeled at 10%). Vetting tenants carefully is critical.
- Income disruption — Option A's lower fixed costs provide more short-term flexibility in a job-loss scenario.